

54, 22, 25, 23, 36, 43, 7, 43, 25, 47, 24, 54, 44

$$\text{Mean} = 34.38$$

a)

x	$x - \bar{x}$	$(x - \bar{x})^2$
54	19.62	384.94
22	-12.38	153.26
25	9.62	87.98
23	-11.38	128.59
36	1.62	2.6244
43	8.62	74.3044
7	-27.38	749.6644
43	8.62	74.3044
25	-9.62	87.9844
47	12.62	159.26
25	-10.38	107.74
54	19.62	384.94
44	9.62	92.544
		$\Sigma = 2488.11$

$$\sigma = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n-1}} \quad (\text{sample})$$

$$\sigma = \sqrt{\frac{2488.11}{12}} = 14.399 \approx 14.4$$

$$C_v = \frac{s}{\bar{x}} \times 100 = \frac{14.4}{34.38} \times 100$$

$$C_v = 41.88\%$$

arranged dataset =

7, 22, 23, 24, 25, 25, 36, 43, 43, 44, 47, 54, 54

b) D_4 & P_{31}

$$D_4 = \frac{4}{10} (n+1)^{th} = \frac{4}{10} (14)^{th} = 5.6^{th} \text{ value}$$

$$\begin{aligned} 5.6^{th} &= 5^{th} + 0.6 (6^{th} - 5^{th}) \\ &= 25 + 0.6 (25 - 25) \end{aligned}$$

$$\boxed{D_4 = 25}$$

$$P_{31} = \frac{31}{100} (n+1)^{th} = \frac{31}{100} (14)^{th} = 4.34^{th} \text{ value}$$

$$= 4^{th} + 0.34 (5^{th} - 4^{th}) = 24 + 0.34 (25 - 24)$$

$$\boxed{P_{31} = 24.34}$$

OR

$$P_{33} = \frac{33}{100} (n+1)^{th} = \frac{33}{100} (14)^{th} = 4.62^{th}$$

$$= 4^{th} + 0.62 (5^{th} - 4^{th}) = 24 + 0.62 (25 - 24)$$

$$\boxed{P_{33} = 24.62}$$

c) Quantiles:

$$\begin{aligned} Q_1 &= \frac{1}{4} (n+1)^{th} = \frac{1}{4} (14)^{th} = 3.5^{th} = 3^{rd} + 0.5 (4^{th} - 3^{rd}) \\ &= 23 + 0.5 (24 - 23) = \boxed{23.5} \end{aligned}$$

$$Q_2 = \frac{2}{4} (14)^{th} = 7^{th} = \boxed{36}$$

$$\begin{aligned} Q_3 &= \frac{3}{4} (14)^{th} = 10.5^{th} = 10^{th} + 0.5 (11^{th} - 10^{th}) \\ &= 44 + 0.5 (47 - 44) = \boxed{45.5} \end{aligned}$$

$$IQR = Q_3 - Q_1 = 45.5 - 23.5 = \boxed{22}$$

d) Outliers:

$$L.F = Q_1 - 1.5(IQR) = -9.5$$

$$U.F = Q_3 + 1.5(IQR) = 78.5$$

So Outliers = None

e) Percentile rank of 36:

$$PR \text{ of } 36 = \frac{\text{no. of values before that}}{\text{Total no. of values}} \times 100$$

$$= \frac{6}{13} \times 100$$

Percentile Rank
of 36

$$= \boxed{46.15\%}$$